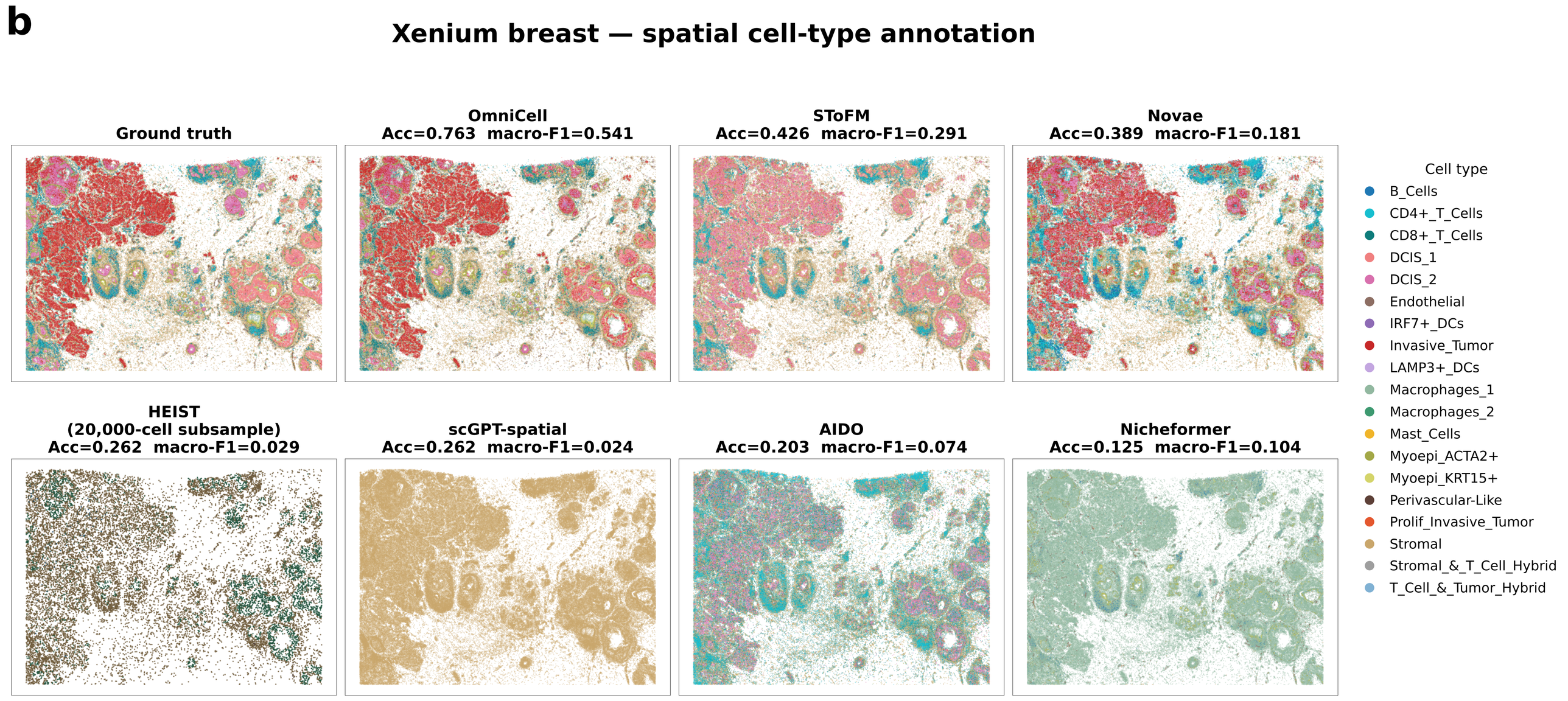

b
Xenium breast — spatial cell-type annotation
Ground truth
OmniCell
Acc=0.763 macro-F1=0.541
SToFM
Acc=0.426 macro-F1=0.291
Novae
Acc=0.389 macro-F1=0.181
HEIST
(20,000-cell subsample)
Acc=0.262 macro-F1=0.029
scGPT-spatial
Acc=0.262 macro-F1=0.024
AIDO
Acc=0.203 macro-F1=0.074
Nicheformer
Acc=0.125 macro-F1=0.104
Cell type
B_Cells
CD4+_T_Cells
CD8+_T_Cells
DCIS_1
DCIS_2
Endothelial
IRF7+_DCs
Invasive_Tumor
LAMP3+_DCs
Macrophages_1
Macrophages_2
Mast_Cells
Myoepi_ACTA2+
Myoepi_KRT15+
Perivascular-Like
Prolif_Invasive_Tumor
Stromal
Stromal_&_T_Cell_Hybrid
T_Cell_&_Tumor_Hybrid